

Lecture - 7

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

KARL PEARSON'S COEFFICIENT OF CORRELATION

Karl Pearson's method of calculating coefficient of correlation is based on the covariance of the two variables in a series. This method is widely used in practice and the coefficient of correlation is denoted by the symbol “*r*”. If the two variables under study are X and Y, the following formula suggested by Karl Pearson can be used for measuring the degree of relationship of correlation.

$$r = \frac{\text{Covariance}(x, y)}{S.D.(x)S.D.(y)}$$

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$r = \frac{\sum XY}{\sqrt{\sum X^2 \sum Y^2}}, \text{ where } X = x - \bar{x} \text{ and } Y = y - \bar{y}$$

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n(\sum x^2) - (\sum x)^2][n(\sum y^2) - (\sum y)^2]}}$$

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2 \sum(Y - \bar{Y})^2}}$$

where $\bar{X}$ and $\bar{Y}$ are the mean of X and Y variable respectively

$$r = \frac{\sum f(dx)(dy) - \frac{(\sum f dx)(\sum f dy)}{N}}{\sqrt{\sum (f dx)^2 - \frac{(\sum f dx)^2}{N}} \sqrt{\sum (f dy)^2 - \frac{(\sum f dy)^2}{N}}}$$

$$dx = X - A \quad \text{and} \quad dy = Y - A$$

Above different formulas can be used in different situation depending upon the information given in the problem.

Example 1. From following information find the correlation coefficient between advertisement expenses and sales volume using Karl Pearson's coefficient of correlation method.

Firm	1	2	3	4	5	6	7	8	9	10
Advertisement Exp. (Rs. In Lakhs)	11	13	14	16	16	15	15	14	13	13
Sales Volume (Rs. In Lakhs)	50	50	55	60	65	65	65	60	60	50

Solution: Let us assume that advertisement expenses are variable X and sales volume are variable Y.

Firm	X	Y	x = X - $\bar{X}$	x^2	y = Y - $\bar{Y}$	y^2	xy
1	11	50	-3	9	-8	64	24
2	13	50	-1	1	-8	64	8
3	14	55	0	0	-3	9	0
4	16	60	2	4	2	4	4
5	16	65	2	4	7	49	14

6	15	65	1	1	7	49	7
7	15	65	1	1	7	49	7
8	14	60	0	0	2	4	0
9	13	60	-1	1	2	4	-2
10	13	50	-1	1	-8	64	8
	$\Sigma X = 140$	$\Sigma Y = 580$		$\Sigma X^2 = 22$		$\Sigma Y^2 = 360$	$\Sigma xy = 70$

$$\bar{X} = \frac{\Sigma X}{n} = \frac{140}{10} = 14, \quad \bar{Y} = \frac{\Sigma Y}{n} = \frac{580}{10} = 58,$$

$$r = \frac{\Sigma XY}{\sqrt{\Sigma X^2 \Sigma Y^2}}, \quad \text{where } X = x - \bar{x} \text{ and } Y = y - \bar{y}$$

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}} = \frac{70}{\sqrt{22 * 360}} = \frac{70}{88.9944} = 0.7866$$

Interpretation: From the above calculation it is very clear that there is high degree of positive correlation i.e., **r = 0.7866**, between the two variables. i.e., Increase in advertisement expenses leads to increased sales volume.

Example 2: Find the correlation coefficient between age and playing habits of the following students using Karl Pearson's coefficient of correlation method.

Age	15	16	17	18	19	20
Number of students	250	200	150	120	100	80
Regular Players	200	150	90	48	30	12

Solution: To find the correlation between age and playing habits of the students, we need to compute the percentages of students who are having the playing habit.

Percentage of playing habits = No. of Regular Players / Total No. of Students * 100

Now let us assume that ages of the students are variable X and percentages of playing habits are variable Y.

Calculation of Karl Pearson's coefficient of correlation

Age (X)	No of Students	Regular Players	Percentage of Playing Habits (Y)	$X - \bar{X}$	$(X - \bar{X})^2$	$Y - \bar{Y}$	$(Y - \bar{Y})^2$	$(X - \bar{X})(Y - \bar{Y})$
15	250	200	80	-2.5	6.25	30	900	-75
16	200	150	75	-1.5	2.25	25	625	-37.5
17	150	90	60	-0.5	0.25	10	100	-5
18	120	48	40	0.5	0.25	-10	100	-5
19	100	30	30	1.5	2.25	-20	400	-30
20	80	12	15	2.5	6.25	-35	1225	-87.5
105			300		17.5		3350	-240
ΣX			ΣY		Σx^2		Σy^2	Σxy

$$\bar{X} = \frac{\Sigma X}{n} = \frac{105}{6} = 17.5, \quad \bar{Y} = \frac{\Sigma Y}{n} = \frac{300}{6} = 50,$$

$$r = \frac{\Sigma(X - \bar{X})(Y - \bar{Y})}{\sqrt{\Sigma(X - \bar{X})^2 \Sigma(Y - \bar{Y})^2}} = -\frac{240}{\sqrt{17.5 * 3350}} = -\frac{240}{242.126} = -0.9912$$

Interpretation: From the above calculation it is very clear that there is high degree of negative correlation i.e., $r = -0.9912$, between the two variables of age and playing habits. i.e., Playing habits among students decreases when their age increases.